

Pull Up Resistor Configuration.

For iCE40UL device, user can configure the internal pull up resistor strength of an IO to a predefined resistor value via attribute settings. By default, all the IO's are weakly pulled up by 100K internal resistor. The PULLUP_RESISTOR attribute is effective only when PULLUP parameter is set to 1.

Synthesis Attribute Syntax:

```
/* synthesis PULLUP_RESISTOR = "3P3K" */
```

Resistor Value:

Resistor Value	Description.
"3P3K"	Pull up resistor level is 3.3K.
"6P8K"	Pull up resistor level is 6.8K.
"10K"	Pull up resistor level is 10K.
"100K"	Pull up resistor level is 100K. (Default)

Note: PULLUP_RESISTOR attribute is supported only for iCE40UL devices.

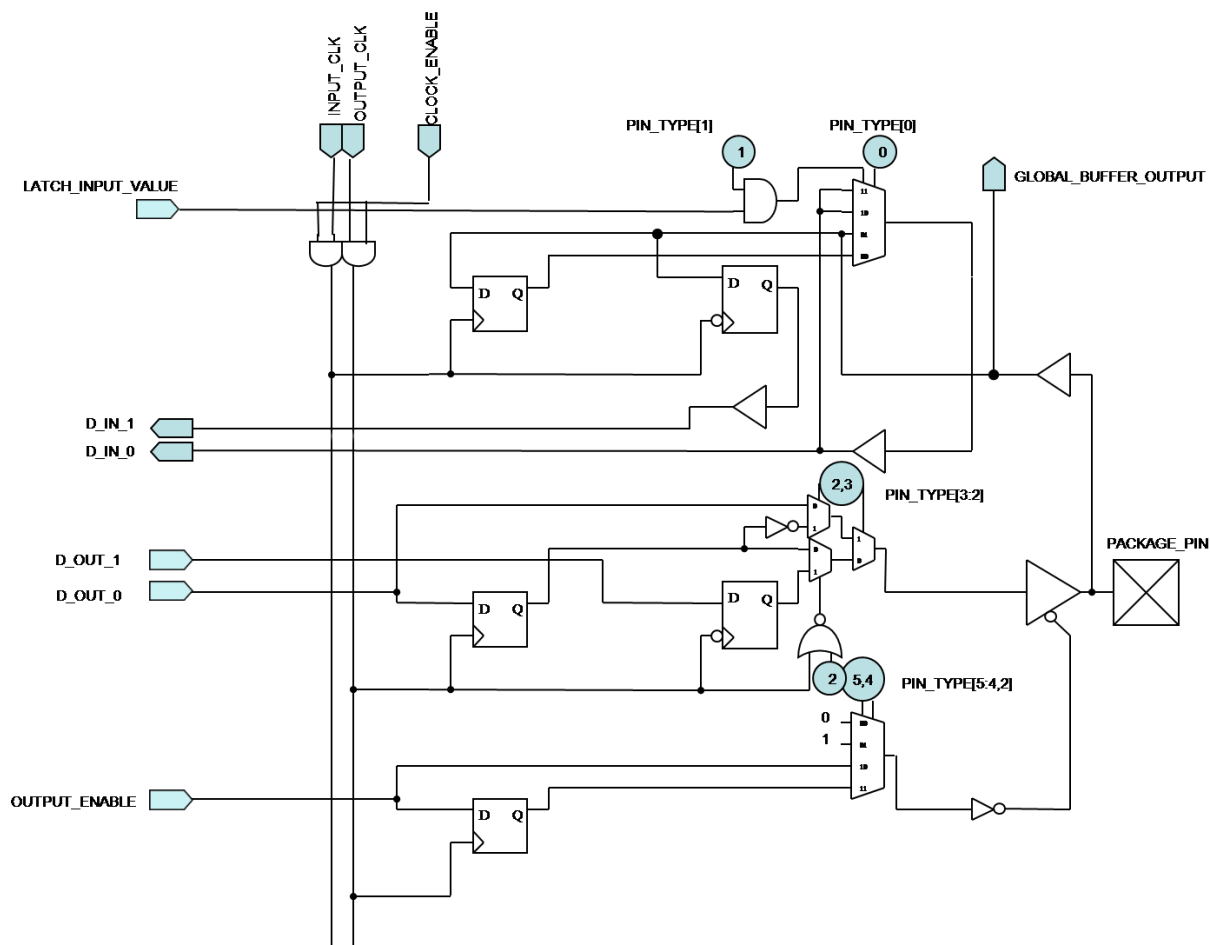
Verilog Instantiation

```
SB_IO      IO_PIN_INST
(
  .PACKAGE_PIN (Package_Pin),           // User's Pin signal name
  .LATCH_INPUT_VALUE (latch_input_value), // Latches/holds the Input value
  .CLOCK_ENABLE (clock_enable),          // Clock Enable common to input and
                                          // output clock
  .INPUT_CLK (input_clk),                // Clock for the input registers
  .OUTPUT_CLK (output_clk),              // Clock for the output registers
  .OUTPUT_ENABLE (output_enable),        // Output Pin Tristate/Enable
                                          // control
  .D_OUT_0 (d_out_0),                    // Data 0 - out to Pin/Rising clk
                                          // edge
  .D_OUT_1 (d_out_1),                    // Data 1 - out to Pin/Falling clk
                                          // edge
  .D_IN_0 (d_in_0),                      // Data 0 - Pin input/Rising clk
                                          // edge
  .D_IN_1 (d_in_1)                       // Data 1 - Pin input/Falling clk
                                          // edge
) /* synthesis DRIVE_STRENGTH= x2 */;

defparam IO_PIN_INST.PIN_TYPE = 6'b000000;
// See Input and Output Pin Function Tables.
// Default value of PIN_TYPE = 6'000000 i.e.
// an input pad, with the input signal
// registered.
defparam IO_PIN_INST.PULLUP = 1'b0;
// By default, the IO will have NO pull up.
// This parameter is used only on bank 0, 1,
// and 2. Ignored when it is placed at bank 3
defparam IO_PIN_INST.NEG_TRIGGER = 1'b0;
// Specify the polarity of all FFs in the IO to
// be falling edge when NEG_TRIGGER = 1.
// Default is rising edge.
defparam IO_PIN_INST.IO_STANDARD = "SB_LVCMOS";
// Other IO standards are supported in bank 3
// only: SB_SSTL2_CLASS_2, SB_SSTL2_CLASS_1,
// SB_SSTL18_FULL, SB_SSTL18_HALF, SB_MDDR10,
// SB_MDDR8, SB_MDDR4, SB_MDDR2 etc.
```

Global Buffer Primitives

SB_GB_IO



Default Signal Values

The iCEcube2 software assigns the logic '0' value to all unconnected input ports except for CLOCK_ENABLE.

Note that explicitly connecting a logic '1' value to port CLOCK_ENABLE will result in a non-optimal implementation, since an extra LUT will be used to generate the Logic '1'. If the user's intention is to keep the Input and Output registers always enabled, it is recommended that port CLOCK_ENABLE be left unconnected.

Verilog Instantiation

```
SB_GB_IO My_Clock_Buffer_Package_Pin (    // A users external Clock reference
                                           pin
    .PACKAGE_PIN (Package_Pin),           // User's Pin signal name
    .LATCH_INPUT_VALUE (latch_input_value), // Latches/holds the Input value
    .CLOCK_ENABLE (clock_enable),         // Clock Enable common to input and
                                           // output clock
```

```

.INPUT_CLK (input_clk),           // Clock for the input registers
.OUTPUT_CLK (output_clk),         // Clock for the output registers
.OUTPUT_ENABLE (output_enable),   // Output Pin Tristate/Enable
                                  // control
.D_OUT_0 (d_out_0),              // Data 0 - out to Pin/Rising clk
                                  // edge
.D_OUT_1 (d_out_1),              // Data 1 - out to Pin/Falling clk
                                  // edge
.D_IN_0 (d_in_0),                // Data 0 - Pin input/Rising clk
                                  // edge
.D_IN_1 (d_in_1)                 // Data 1 - Pin input/Falling clk
                                  // edge

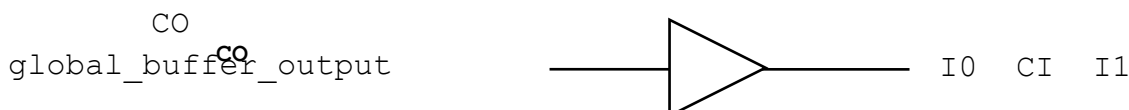
.GLOBAL_BUFFER_OUTPUT (Global_Buffered_User_Clock)
                                  // Example use - clock buffer
                                  //driven from the input pin
);

defparam My_Clock_Buffer_Package_Pin.PIN_TYPE = 6'b000000;
                                  // See Input and Output Pin Function Tables.
                                  // Default value of PIN_TYPE = 6'000000 i.e.
                                  // an input pad, with the input signal
                                  // registered

```

Note that this primitive is a superset of the SB_IO primitive, and includes the connectivity to drive a Global Buffer. For example SB_GB_IO pins are likely to be used for external Clocks.

SB_GB Primitive



Verilog Instantiation

```

SB_GB My_Global_Buffer_i (        //Required for a user's internally generated
                                  //FPGA signal that is heavily loaded and
                                  //requires global buffering. For example, a
                                  //user's logic-generated clock.

.USER_SIGNAL_TO_GLOBAL_BUFFER (Users_internal_Clk),
.GLOBAL_BUFFER_OUTPUT ( Global_Buffered_User_Signal)

);

```

PLL Primitives

The Phase Lock Loop (PLL) function is offered as a feature in certain iCE device packages.

It is strongly recommended that the configuration of the PLL primitives be accomplished through the use of the PLL Configuration tool that is offered as part of the iCEcube2 software.

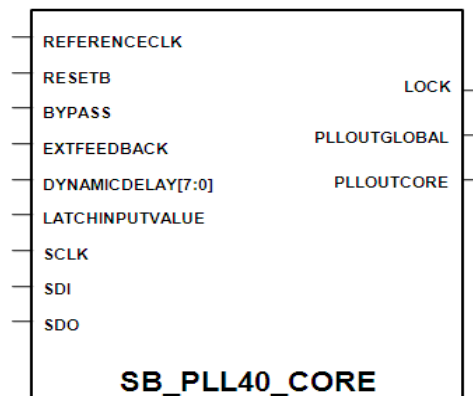
iCE40 PLL Primitives

There are 5 primitives that represent the PLL function in the iCEcube2 software viz. SB_PLL40_CORE, SB_PLL40_PAD, SB_PLL40_2_PAD, SB_PLL40_2F_CORE and SB_PLL40_2F_PAD for the ice40 device family. A short description of each primitive and its ports/parameters is provided in the following sections.

It is strongly recommended that the configuration of the PLL primitives be accomplished through the use of the PLL Configuration tool that is offered as part of the iCEcube2 software.

SB_PLL40_CORE

The SB_PLL40_CORE primitive should be used when the source clock of the PLL is driven by FPGA routing i.e. when the PLL source clock originates on the FPGA or is driven by an input pad that is not in the bottom IO bank (IO Bank 2).



Ports

REFERENCECLK: PLL source clock that serves as the input to the SB_PLL40_CORE primitive.

PLLOUTGLOBAL: Output clock generated by the PLL, drives a global clock network on the FPGA.

PLLOUTCORE: Output clock generated by the PLL, drives regular FPGA routing. The frequency generated on this output is the same as the frequency of the clock signal generated on the PLLOUTGLOBAL port.

LOCK: Output port, when HIGH, indicates that the signal on PLLOUTGLOBAL/PLLOUTCORE is locked to the PLL source on REFERENCECLK.

EXTFEEDBACK: External feedback input to PLL. Enabled when the FEEDBACK_PATH parameter is set to EXTERNAL.

DYNAMICDELAY: 7 bit input bus that enables dynamic control of the delay contributed by the Fine Delay Adjust Block. The Fine Delay Adjust Block is used when there is a need to adjust the phase alignment of PLLOUTGLOBAL/PLLOUTCORE with respect to REFERENCECLK. The DYNAMICDELAY port controls are enabled when the DELAY_ADJUSTMENT_MODE parameter is set to DYNAMIC.

RESETB: Active low input that asynchronously resets the PLL.

BYPASS: Input signal, when asserted, connects the signal on REFERENCECLK to PLLOUTCORE/PLLOUTGLOBAL pins.

LATCHINPUTVALUE: Active high input, when enabled, forces the PLL into low-power mode. The PLLOUTGLOBAL/PLLOUTCORE pins are held static at their last value. This function is enabled when the parameter ENABLE_ICEGATE is set to '1'.

SCLK, *SDI*, *SDO*: These pins are used only for internal testing purposes, and need not be instantiated by users.

Parameters

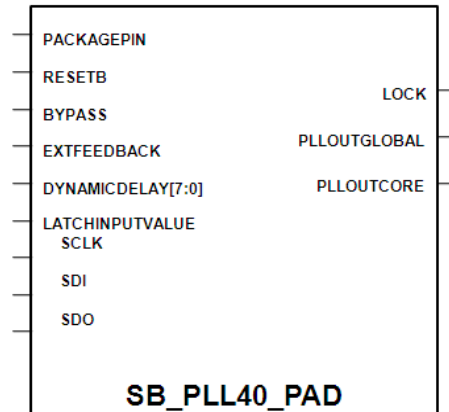
The SB_PLL40_CORE primitive requires configuration through the specification of the following parameters. It is strongly recommended that the configuration of the PLL primitives be accomplished through the use of the PLL Configuration tool that is offered as part of the iCEcube2 software.

Parameter Name	Description	Parameter Value	Description
FEEDBACK_PATH	Selects the feedback path to the PLL	SIMPLE	Feedback is internal to the PLL, directly from VCO
		DELAY	Feedback is internal to the PLL, through the Fine Delay Adjust Block
		PHASE_AND_DELAY	Feedback is internal to the PLL, through the Phase Shifter and the Fine Delay Adjust Block
		EXTERNAL	Feedback path is external to the PLL, and connects to EXTFEEDBACK pin. Also uses the Fine Delay Adjust Block.
DELAY_ADJUSTMENT_MODE_FEEDBACK	Selects the mode for the Fine Delay Adjust block in the feedback path	FIXED	Delay of the Fine Delay Adjust Block is fixed, the value is specified by the FDA_FEEDBACK parameter setting
		DYNAMIC	Delay of Fine Delay Adjust Block is determined by the signal value at the DYNAMICDELAY[3:0] pins
FDA_FEEDBACK	Sets a constant value for the Fine Delay Adjust Block in the feedback path	0, 1,...,15	The PLLOUTGLOBAL & PLLOUTCORE signals are delay compensated by $(n+1)*150$ ps, where $n = \text{FDA_FEEDBACK}$ only if the setting of the DELAY_ADJUSTMENT_MODE_FEEDBACK is FIXED.
DELAY_ADJUSTMENT_MODE_RELATIVE	Selects the mode for the Fine Delay Adjust block	FIXED	Delay of the Fine Delay Adjust Block is fixed, the value is specified by the FDA_RELATIVE parameter setting
		DYNAMIC	Delay of Fine Delay Adjust Block is determined by the signal value at the DYNAMICDELAY[7:4] pins
FDA_RELATIVE	Sets a constant value for the Fine Delay Adjust Block	0, 1,...,15	The PLLOUTGLOBALA & PLLOUTCOREA signals are additionally delayed by $(n+1)*150$ ps, where $n = \text{FDA_RELATIVE}$. Used if DELAY_ADJUSTMENT_MODE_RELATIVE is "FIXED".
SHIFTREG_DIV_MODE	Selects shift register configuration	0,1	Used when FEEDBACK_PATH is "PHASE_AND_DELAY". 0→Divide by 4 1→Divide by 7
PLLOUT_SELECT	Selects the signal to be output at the PLLOUTCORE and PLLOUTGLOBAL ports	SHIFTREG_0deg	0° phase shift only if the setting of FEEDBACK_PATH is "PHASE_AND_DELAY"
		SHIFTREG_90deg	90° phase shift only if the setting of FEEDBACK_PATH is "PHASE_AND_DELAY" and SHIFTREG_DIV_MODE=0
		GENCLK	The internally generated PLL frequency will be output without any phase shift.
		GENCLK_HALF	The internally generated PLL frequency will be divided by 2 and then output. No phase shift.
DIVR	REFERENCECLK divider	0,1,2,...,15	These parameters are used to

DIVF	Feedback divider	0,1,...,63	control the output frequency, depending on the FEEDBACK_PATH setting.
DIVQ	VCO Divider	1,2,...,6	
FILTER_RANGE	PLL Filter Range	0,1,...,7	
EXTERNAL_DIVIDE_FACTOR	Divide-by factor of a divider in external feedback path	User specified value. Default 1	Specified only when there is a user-implemented divider in the external feedback path.
ENABLE_ICEGATE	Enables the PLL power-down control	0	Power-down control disabled
		1	Power-down controlled by LATCHINPUTVALUE input

SB_PLL40_PAD

The SB_PLL40_PAD primitive should be used when the source clock of the PLL is driven by an input pad that is located in the bottom IO bank (IO Bank 2) or the top IO bank (IO Bank 0), and the source clock is not required inside the FPGA.



Ports

PACKAGEPIN: PLL REFERENCE CLOCK pin that serves as the input to the SB_PLL40_PAD primitive.

PLLOUTGLOBAL: Output clock generated by the PLL, drives a global clock network on the FPGA.

PLLOUTCORE: Output clock generated by the PLL, drives regular FPGA routing. The frequency generated on this output is the same as the frequency of the clock signal generated on the PLLOUTGLOBAL port.

LOCK: Output port, when HIGH, indicates that the signal on PLLOUTGLOBAL/PLLOUTCORE is locked to the PLL source on PACKAGEPIN.

EXTFEEDBACK: External feedback input to PLL. Enabled when the FEEDBACK_PATH parameter is set to EXTERNAL.

DYNAMICDELAY: 7 bit input bus that enables dynamic control of the delay contributed by the Fine Delay Adjust Block. The Fine Delay Adjust Block is used when there is a need to adjust the phase alignment of PLLOUTGLOBAL/PLLOUTCORE with respect to PACKAGEPIN. The DYNAMICDELAY port controls are enabled when the DELAY_ADJUSTMENT_MODE parameter is set to DYNAMIC.

RESETB: Active low input that asynchronously resets the PLL.

BYPASS: Input signal, when asserted, connects the signal on PACKAGEPIN to PLLOUTCORE/PLLOUTGLOBAL pins.

LATCHINPUTVALUE: Active high input, when enabled, forces the PLL into low-power mode. The PLLOUTGLOBAL/PLLOUTCORE pins are held static at their last value. This function is enabled when the parameter ENABLE_ICEGATE is set to '1'.

SCLK, SDI, SDO: These pins are used only for internal testing purposes, and need not be instantiated by users.

Parameters

The SB_PLL40_PAD primitive requires configuration through the specification of the following parameters. It is strongly recommended that the configuration of the PLL primitives be accomplished through the use of the PLL Configuration tool that is offered as part of the iCEcube2 software.

Parameter Name	Description	Parameter Value	Description
FEEDBACK_PATH	Selects the feedback path to the PLL	SIMPLE	Feedback is internal to the PLL, directly from VCO
		DELAY	Feedback is internal to the PLL, through the Fine Delay Adjust Block
		PHASE_AND_DELAY	Feedback is internal to the PLL, through the Phase Shifter and the Fine Delay Adjust Block
		EXTERNAL	Feedback path is external to the PLL, and connects to EXTFEEDBACK pin. Also uses the Fine Delay Adjust Block.
DELAY_ADJUSTMENT_MODE_FEEDBACK	Selects the mode for the Fine Delay Adjust block in the feedback path	FIXED	Delay of the Fine Delay Adjust Block is fixed, the value is specified by the FDA_FEEDBACK parameter setting
		DYNAMIC	Delay of Fine Delay Adjust Block is determined by the signal value at the DYNAMICDELAY [3:0] pins
FDA_FEEDBACK	Sets a constant value for the Fine Delay Adjust Block in the feedback path	0, 1,...,15	The PLLOUTGLOBAL & PLLOUTCORE signals are delay compensated by $(n+1)*150$ ps, where $n = \text{FDA_FEEDBACK}$ only if the setting of the DELAY_ADJUSTMENT_MODE_FEEDBACK is FIXED.
DELAY_ADJUSTMENT_MODE_RELATIVE	Selects the mode for the Fine Delay Adjust block	FIXED	Delay of the Fine Delay Adjust Block is fixed, the value is specified by the FDA_RELATIVE parameter setting
		DYNAMIC	Delay of Fine Delay Adjust Block is determined by the signal value at the DYNAMICDELAY[7:4] pins
FDA_RELATIVE	Sets a constant value for the Fine Delay Adjust Block	0, 1,...,15	The PLLOUTGLOBALA & PLLOUTCOREA signals are additionally delayed by $(n+1)*150$ ps, where $n = \text{FDA_RELATIVE}$. Used if DELAY_ADJUSTMENT_MODE_RELATIVE is "FIXED".
SHIFTREG_DIV_MODE	Selects shift register configuration	0,1	Used when FEEDBACK_PATH is "PHASE_AND_DELAY". 0→Divide by 4 1→Divide by 7
PLLOUT_SELECT	Selects the signal to be output at the PLLOUTCORE and PLLOUTGLOBAL ports	SHIFTREG_0deg	0° phase shift only if the setting of FEEDBACK_PATH is "PHASE_AND_DELAY"
		SHIFTREG_90deg	90° phase shift only if the setting of FEEDBACK_PATH is "PHASE_AND_DELAY" and SHIFTREG_DIV_MODE=0
		GENCLK	The internally generated PLL frequency will be output without any phase shift.
		GENCLK_HALF	The internally generated PLL frequency will be divided by 2 and then output. No phase shift.
DIVR	REFERENCECLK divider	0,1,2,...,15	These parameters are used to